

E-Commerce Product Analytics & Conversion Funnel Analysis

A Product Analyst's Diagnosis of Where, Why, and What to Fix

1,400,976	1,683,380	12,175	0.72%
Total Visitors	Total Sessions	Total Purchases	Visitor → Purchase Rate

Dataset: RetailRocket E-Commerce Events (Kaggle) — 2.75M raw events, 4.5 months of activity

Tech Stack: Python (pandas) · PostgreSQL · Streamlit · Plotly

Prepared as a Product Analytics Case Study

1. Executive Summary

The company observed high website traffic paired with low conversion and high cart abandonment. This analysis traced the full user journey — from product view through purchase — across 1.4 million visitors to identify exactly where value is being lost and why.

The core finding: the funnel does not leak gradually. It collapses almost entirely at the very first step. Only 1.98% of product views ever result in an Add to Cart action — by far the largest drop in the funnel. Once a visitor commits to adding an item to their cart, they go on to complete the purchase 26.34% of the time, a rate over 13x higher than the initial view-to-cart conversion. This means checkout itself is comparatively healthy; the real bottleneck is product-page relevance and first-visit engagement, not friction at checkout.

This is reinforced by session-level and cohort data: 79.57% of sessions end after a single event, 97.45% of visitors never add anything to cart, and every weekly cohort shows retention collapsing to 3–4% by week one. Repeat purchase rate holds steady around 0.7% across five months of data, indicating a structural, ongoing issue rather than a recent decline.

Top Recommendations

- Audit high-traffic, zero-purchase categories (e.g. categories 1007, 697, 173, 561, 741) for pricing, imagery, and stock accuracy issues
- Prioritize product-page and search relevance improvements over checkout optimization
- Invest in first-session experience, since nearly all value capture happens in session one
- Study top-converting categories (1685 at 5.86%, 340 at 5.14%) to replicate what is working elsewhere in the catalog

2. Business Problem

As Product Analyst, the mandate was to diagnose three linked symptoms: high traffic, low conversion, and high cart abandonment. Management needed clear answers to four questions:

- Where are users dropping in the funnel?
- Why are they dropping?
- Which users convert the most?
- What product changes should be implemented?

The objective was to identify the highest-leverage fix — without assuming more acquisition spend was the answer, since traffic was already high.

3. Dataset

The analysis uses the **RetailRocket E-Commerce Events** dataset, chosen over alternatives (GA4 sample data, Olist, Instacart) for its event-level granularity: individual view, add-to-cart, and transaction events tied to visitor and item identifiers.

File	Grain	Rows (approx.)
events.csv	One row per user event	2,756,101
item_properties (parts 1 & 2)	One row per item property change	~20,275,902
category_tree.csv	One row per category	~1,700

Known Data Limitations

- No price field — AOV and Revenue KPIs could not be directly calculated
- No device or country fields — geographic and device segmentation were out of scope for this dataset
- Bot/abnormal traffic (~6,600 visitors, 367K events) was filtered using a 99.5th-percentile event-count threshold prior to analysis

4. Methodology

4.1 Data Cleaning Pipeline

- Type correction: timestamps converted to datetime; IDs cast as strings to prevent accidental numeric operations
- Deduplication of exact-match event rows
- Bot traffic removed using a 99.5th-percentile event-count-per-visitor threshold
- Chronological sorting per visitor to support session logic
- Session generation using a 30-minute inactivity rule
- Item properties pivoted from long (EAV) to wide format, keeping the latest categoryid and availability value per item
- Final join of events, item properties, and category tree into a single analysis-ready table

4.2 Database Design

Cleaned data was loaded into a normalized PostgreSQL star schema: a central **events** fact table referencing **visitors**, **items**, **categories**, and **sessions** dimension tables, indexed on visitor, session, item, event type, and timestamp for query performance across the ~2.4M-row event table.

4.3 A Data Quality Fix Worth Noting

During analysis, category IDs initially appeared limited to only two values (0 and 1) across the entire dataset — a red flag, since a real catalog should show hundreds of distinct categories. Root-cause investigation traced this to a column-alignment issue during the CSV-to-database import. The itemid-to-category mapping was rebuilt directly from the source data, restoring the full range of category IDs (with realistic, varying item counts per category) before any category-level conclusions were drawn. This is called out here because the Worst-Performing Categories finding in Section 6 depends on this corrected mapping.

5. Exploratory Data Analysis

Metric	Value
Unique visitors	1,400,976
Unique sessions	1,683,380
Unique items interacted with	226,791
Avg. events per session	1.42
Avg. session duration	1.30 minutes

Bounce rate (single-event sessions)	79.57%
Peak traffic day	Tuesday
Peak traffic hour	8–9 PM
Peak purchase hour	5 PM
Items viewed but never purchased	219,051 (96.6%)

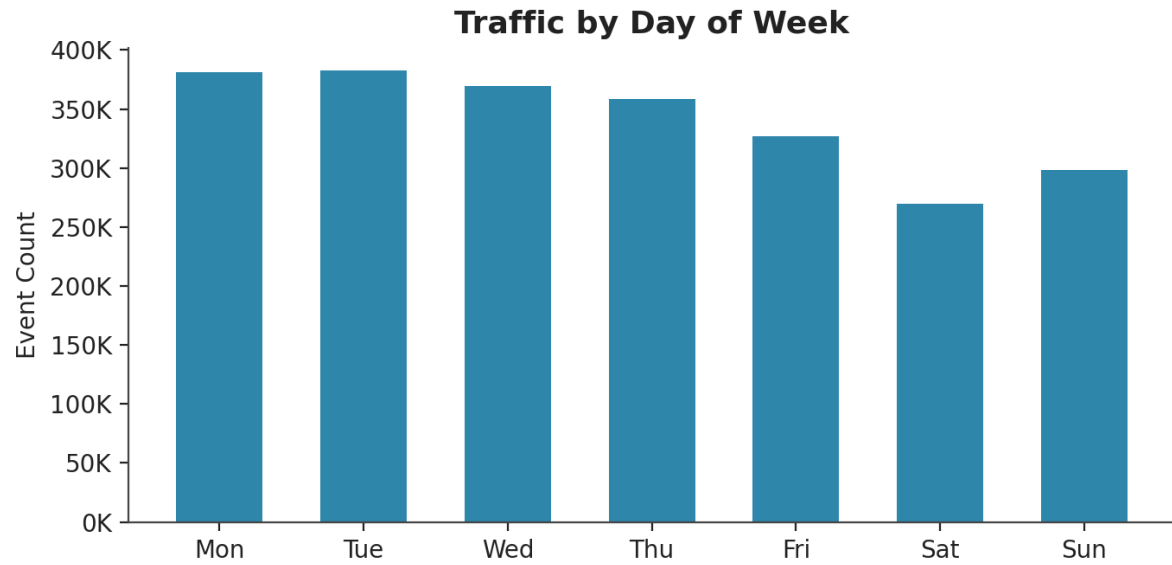


Figure 1. Weekday traffic peaks Monday–Tuesday and dips through the weekend.

Thirty EDA questions were answered in total, spanning visitor volume, session behavior, time patterns, product performance, category performance, user segmentation, and cart/order behavior. The full question set and results are documented in the project repository.

6. Funnel Analysis

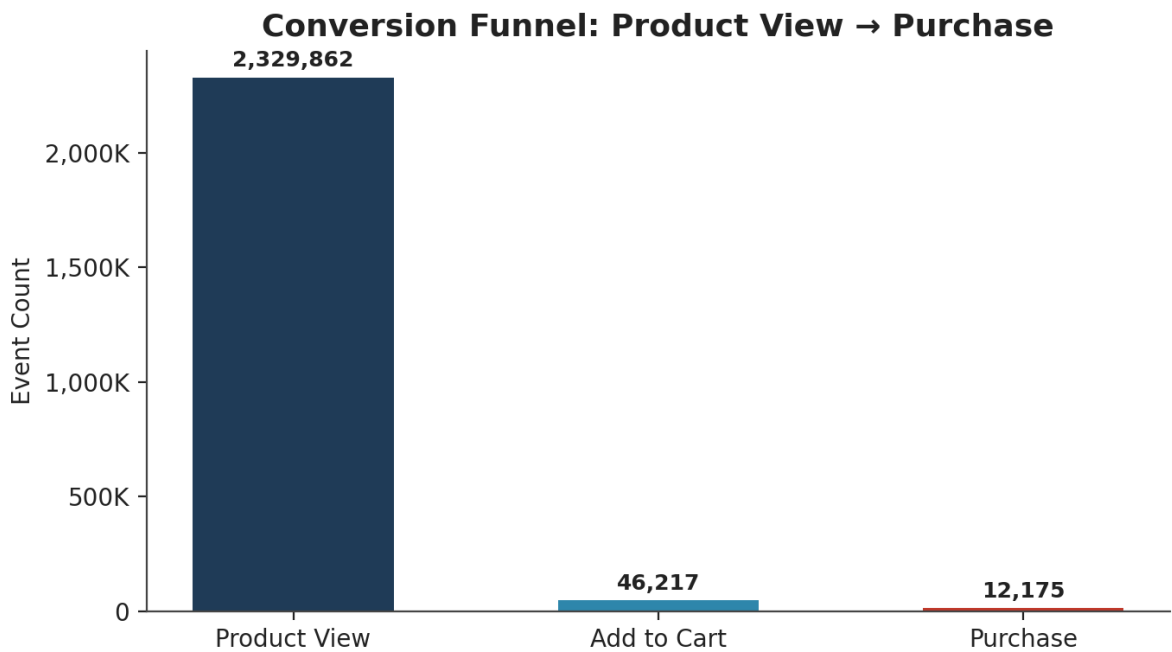


Figure 2. The drop between Product View and Add to Cart dwarfs every other stage transition.

Stage	Count	Conversion from Previous	Drop-off
Product View	2,329,862	—	—
Add to Cart	46,217	1.98%	98.02%
Purchase	12,175	26.34%	73.66%

Overall View → Purchase conversion: 0.52%

Where's the Largest Drop?

The View-to-Cart transition is the overwhelming bottleneck. Once a visitor adds an item to cart, they convert to purchase over 13 times more often than the base rate of product-page visitors adding to cart at all. Checkout, in other words, is not the problem — getting visitors to engage past a single product view is.

Category-Level Evidence

Category ID	Views	Purchases
1007	4,512	0
697	3,257	0
173	2,414	0

561	2,172	0
741	2,098	0

These categories combine substantial traffic (1,200–4,500+ views each) with **zero** purchases — a systemic issue rather than statistical noise, and the highest-leverage starting point for a merchandising audit.

By contrast, the best-converting categories (1685 at 5.86%, 340 at 5.14%) prove that high conversion is achievable within this catalog when product/category fit is right — these are worth studying as a template for the underperformers.

7. User Segmentation & Cohort Analysis

Visitor Segments (Behavior-Based)

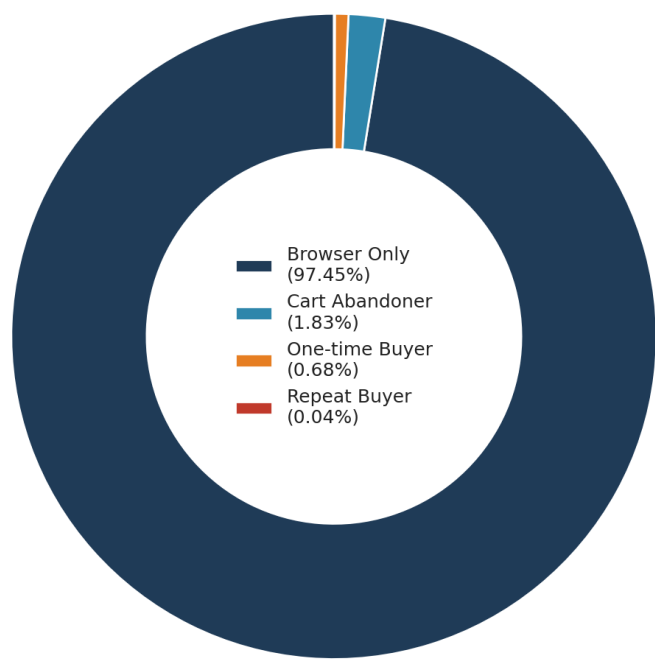


Figure 3. Behavior-based visitor segmentation.

Segment	% of Visitors	Avg. Sessions	Avg. Views	Avg. Cart Adds
Repeat Buyer	0.04%	2.72	6.13	2.14
One-time Buyer	0.68%	1.96	4.16	1.15
Cart Abandoner	1.83%	1.61	3.36	1.33
Browser Only	97.45%	1.19	1.61	0.00

There is a clean engagement gradient across segments: Repeat Buyers average 2.72 sessions and 6.13 views versus Browser Only visitors' 1.19 sessions and 1.61 views. This confirms engagement depth, not just presence, is what separates buyers from the vast non-converting majority.

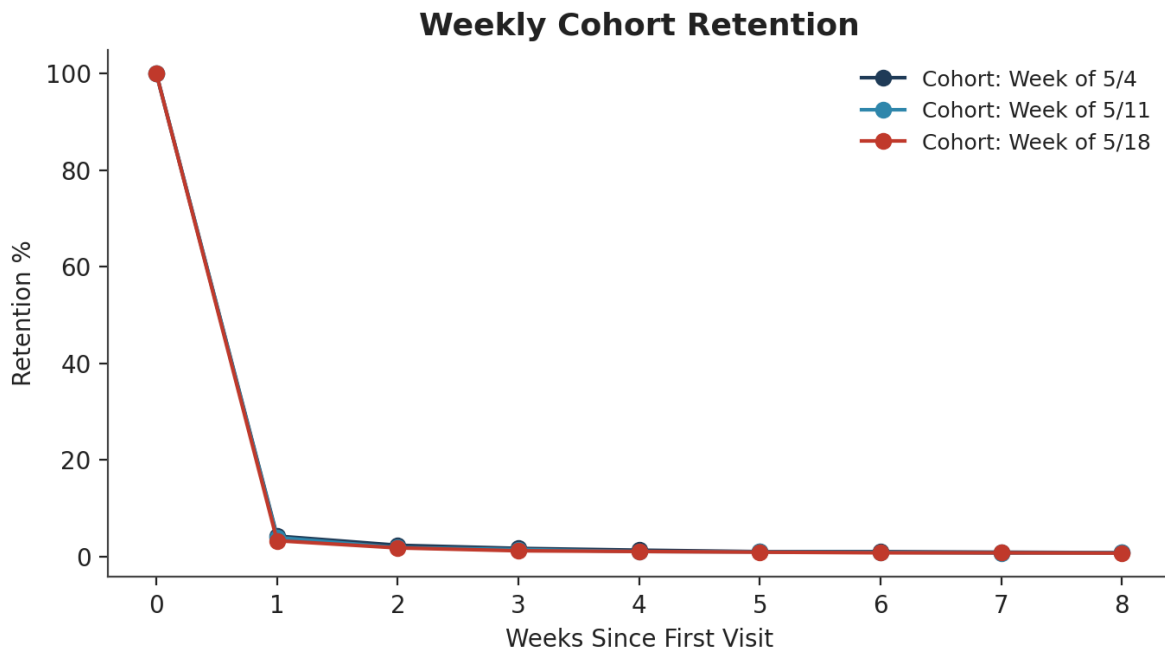


Figure 4. Every weekly cohort follows the same collapse: ~100% → ~3-4% retention by Week 1.

Repeat purchase rate held steady in the 0.55%–0.88% range across 21 weekly cohorts spanning the full ~5-month dataset, with no meaningful upward or downward trend. This indicates the conversion problem is **structural and ongoing**, not a recent regression or a seasonal dip — which means the fix needs to address root cause, not a one-off campaign or promotional issue.

8. A/B Testing Design

Test 1: Product Page Relevance (Recommended First)

Hypothesis: Adding social proof and improving image quality near the Add-to-Cart button will increase View → Cart conversion, since the data shows a relevance/trust gap at the product page rather than at checkout.

- Primary metric: View → Add to Cart conversion rate (baseline 1.98%)
- Minimum detectable effect: 0.3 percentage points
- With ~2.3M product views available, this test reaches significance in days, not months
- Statistical test: two-proportion z-test, segmented by the worst-performing categories identified in Section 6

Test 2: Checkout Redesign (As Originally Scoped)

Hypothesis: A simplified checkout flow (fewer steps, guest checkout, progress indicator) will reduce cart abandonment, assuming friction — not price or trust — is the primary driver of drop-off at this stage.

- Primary metric: Cart Abandonment Rate (baseline 73.66%)

- Minimum detectable effect: 2 percentage points
- Required sample: ~18,800 total users across both variants, achievable in ~2 months of live traffic
- Statistical test: two-proportion z-test on Cart → Purchase rate

Recommended sequencing: Run Test 1 first — it targets the larger, proven leak, requires far less time to reach significance, and its results can inform whether Test 2 is worth prioritizing at all.

9. Product Recommendations

Finding	Recommendation
98% of views never reach cart	Audit zero-purchase categories for pricing, imagery, stock accuracy; improve search/reco
79.6% single-event bounce	Review traffic source quality; test product-page relevance improvements before checkout
Week-1 retention only 3–4%	Invest in first-session experience: search quality, personalization, page load speed
Checkout is comparatively healthy (26.3%)	Do not over-invest in checkout optimization; resources are better spent upstream

10. Future Work

- Incorporate price/revenue data (via a supplementary dataset or estimation model) to quantify the dollar impact of the View-to-Cart leak
- Add device and geographic data (e.g. via a GA4 sample dataset overlay) to segment findings by platform and region
- Run the two proposed A/B tests and feed results back into the product roadmap
- Build category-level root-cause analysis (pricing benchmarks, image quality audits) for the identified zero-purchase categories
- Extend cohort analysis to monthly granularity and track post-fix retention curves once product changes ship

End of report. Dataset: RetailRocket E-Commerce Events (Kaggle). Prepared as a product analytics case study.